Grade 6 Practice

Standard 6.RP.1

Computation

1. Write the ratio of 3 cats to 5 dogs. ___ **3:5**

2. Write the ratio of 4 red to 6 blue in simplest form. ___ **2:3**

3. Write the ratio of 8 to 12 in simplest form. ___ **2:3**

4. Write the ratio of 10 apples to 4 oranges in simplest form. ___ **5:2**

Word problems

5. A class has 12 boys and 18 girls. What is the ratio of boys to girls in simplest form?
Answer: 2:3

6. Sam makes 9 of 12 free throws. What is the ratio of made shots to missed shots in simplest form?
Answer: 3:1

Grade 6 Practice

Standard 6.RP.2

Computation

1. A car goes 150 miles in 3 hours. What is its unit rate in miles per hour? ___ **50**

2. 12 eggs are shared by 4 people. How many eggs per person? ___ **3**

3. 4 pens cost $6. What is the price per pen? ___ **$1.50**

4. A hiker goes 24 km in 3 hours. What is the rate in km per hour? ___ **8**

Word problems

5. Ana types 210 words in 5 minutes. How many words per minute is that?
Answer: 42 words per minute

6. A printer prints 96 pages in 8 minutes. How many pages per minute is that?
Answer: 12 pages per minute

Grade 6 Practice

Standard 6.RP.3

Computation

1. The ratio of boys to girls is 2:3. If there are 6 boys, how many girls are there? ___ **9**

2. A recipe uses 3 cups flour for every 2 cups sugar. How many cups of sugar go with 6

3. 5 pencils cost $10. What is the price of 1 pencil? ___ **$2**

4. On a map, 1 inch equals 50 miles. How many miles is 4 inches? ___ **200**

Word problems

5. Mia reads 3 pages every 5 minutes. How many minutes does it take to read 30 pages?
Answer: 50 minutes

6. A jacket costs $40. It is on sale for 25% off. What is the sale price?
Answer: $30

Grade 6 Practice

Standard 6.NS.1

Computation

1. Solve: 2/3 ÷ 1/6 = ___ **4**

2. Solve: 3/4 ÷ 1/2 = ___ **1 1/2**

3. Solve: 1/2 ÷ 3/4 = ___ **2/3**

4. Solve: 5/6 ÷ 2/3 = ___ **1 1/4**

Word problems

5. A ribbon is 3/4 meter long. How many 1/8-meter pieces can be cut from it?

Answer: 6 pieces

6. How many 2/3-cup servings can be made from 4 cups of rice?

Answer: 6 servings

Grade 6 Practice

Standard 6.NS.2

Computation

1. Solve: 936 ÷ 12 = ___ **78**

2. Solve: 1,428 ÷ 14 = ___ **102**

3. Solve: 2,064 ÷ 24 = ___ **86**

4. Solve: 5,376 ÷ 32 = ___ **168**

Word problems

5. A bus holds 36 students. How many buses are needed for 900 students?
Answer: 25 buses

6. A farm packs 1,250 eggs into cartons that hold 12 eggs. How many full cartons are there?
Answer: 104 full cartons

Grade 6 Practice

Standard 6.NS.3

Computation

1. Solve: 3.25 + 4.8 = ___ **8.05**

2. Solve: 9.4 - 2.67 = ___ **6.73**

3. Solve: 2.5 × 3.2 = ___ **8**

4. Solve: 7.5 ÷ 2.5 = ___ **3**

Word problems

5. Sam buys 3 items that cost $2.45 each. How much does he spend in all?
Answer: $7.35

6. A 12.6-meter rope is cut into 4 equal pieces. How long is each piece?
Answer: 3.15 meters

Grade 6 Practice

Standard 6.NS.4

Computation

1. Find the GCF of 12 and 18. ___ **6**

2. Find the LCM of 4 and 6. ___ **12**

3. Find the GCF of 15 and 20. ___ **5**

4. Find the LCM of 3 and 5. ___ **15**

Word problems

5. Two ribbons are 24 cm and 36 cm long. They are cut into the longest equal pieces possible with none left over. How long is each piece?
Answer: 12 cm

6. Two lights blink every 6 seconds and every 8 seconds. After how many seconds do they blink together?
Answer: 24 seconds

Grade 6 Practice

Standard 6.NS.5

Computation

1. What is the opposite of 5? ___ **-5**

2. Start at -3 and go up 5. Where do you land? ___ **2**

3. Is -2 greater than -7? Write yes or no. **yes**

4. Write a negative number for 40 meters below sea level. ___ **-40**

Word problems

5. A bank balance is -$25. Then $40 is deposited. What is the new balance?
Answer: $15

6. Which temperature is colder, -12°F or -5°F?
Answer: -12°F

Grade 6 Practice

Standard 6.NS.6

Computation

1. What is the opposite of -9? ___ **9**

2. What is the opposite of 0? ___ **0**

3. The point -4 is how many units from 0? ___ **4**

4. In which quadrant is the point (-3, 2)? Write the roman numeral. **II**

Word problems

5. The point (-2, 5) is in which quadrant? Write the roman numeral.
Answer: II

6. The point (-3, -1) is in which quadrant? Write the roman numeral.
Answer: III

Grade 6 Practice

Standard 6.NS.7

Computation

1. What is |-8|? ___ **8**

2. Write < or > : -3 ___ -1 **<**

3. Is |-6| greater than |2|? Write yes or no. **yes**

4. Order from least to greatest: -4, 2, -1. ___ **-4, -1, 2**

Word problems

5. Sam's temperature is -7°C and another is -3°C. Which is colder?
Answer: -7°C

6. Which is less: owing $12 or owing $5?
Answer: owing $12

Grade 6 Practice

Standard 6.NS.8

Computation

1. Find the distance between (2, 3) and (2, 9). ___ **6**

2. Find the distance between (-4, 1) and (3, 1). ___ **7**

3. Find the distance between (0, -5) and (0, 2). ___ **7**

4. Find the distance between (-2, -3) and (-2, 4). ___ **7**

Word problems

5. A store is at (1, 2) and a park is at (1, 10). How many units apart are they?
Answer: 8 units

6. Two points are at (-3, 0) and (5, 0). What is the distance between them?
Answer: 8 units

Grade 6 Practice

Standard 6.EE.1

Computation

1. Evaluate 2^3. ___ **8**

2. Write $5 \times 5 \times 5$ using an exponent. ___ **5^3**

3. Evaluate 4^2. ___ **16**

4. Evaluate 10^3. ___ **1,000**

Word problems

5. A square has side length 6. What is its area? (Area = side^2)
Answer: 36 square units

6. A cube has edges of length 3. What is its volume? (Volume = side^3)
Answer: 27 cubic units

Grade 6 Practice

Standard 6.EE.2

Computation

1. Write an expression for 'a number n plus 7.' ___ **n + 7**

2. Write an expression for 'the product of 4 and x.' ___ **4x**

3. Evaluate 3y - 2 when y = 5. ___ **13**

4. Evaluate 2(a + 4) when a = 1. ___ **10**

Word problems

5. Cookies cost c dollars each. Write an expression for the cost of 6 cookies.
Answer: 6c

6. Tim is t years old. His sister is 4 years older. Write an expression for her age.
Answer: t + 4

Grade 6 Practice

Standard 6.EE.3

Computation

1. Is 2(x + 3) equal to 2x + 6? Write yes or no. **yes**

2. Is 3x + 4 equal to 4 + 3x? Write yes or no. **yes**

3. Simplify 4x + 2x. ___ **6x**

4. Simplify 5 + 2y + 3. ___ **2y + 8**

Word problems

5. Mia says 3(x + 2) = 3x + 2. Is she right? Write yes or no.
Answer: no

6. Simplify 7a - 3a + 2. ___
Answer: 4a + 2

Grade 6 Practice

Standard 6.EE.4

Computation

1. Are 4 + x and x + 4 equivalent? Write yes or no. **yes**

2. Are 2y and y + y equivalent? Write yes or no. **yes**

3. Are 3x and x + 3 equivalent? Write yes or no. **no**

4. Write an expression equivalent to 5(a + 2). ___ **5a + 10**

Word problems

5. Sam says 6 + 2b and 2(3 + b) are equivalent. Is he right? Write yes or no.
Answer: yes

6. Lee says 4 × a and a × 4 are equivalent. Is he right? Write yes or no.
Answer: yes

Grade 6 Practice

Standard 6.EE.5

Computation

1. Is $x = 3$ a solution to $x + 4 = 7$? Write yes or no. **yes**

2. Is $y = 5$ a solution to $2y = 8$? Write yes or no. **no**

3. Is $n = 0$ a solution to $n + 6 = 6$? Write yes or no. **yes**

4. Is $a = 2$ a solution to $3a = 6$? Write yes or no. **yes**

Word problems

5. Is $m = 4$ a solution to $5m - 1 = 18$? Write yes or no.
Answer: no

6. Is $p = 7$ a solution to $p - 3 = 4$? Write yes or no.
Answer: yes

Grade 6 Practice

Standard 6.EE.6

Computation

1. Write an expression for the number of cookies in n boxes, with 12 cookies per box. ___

2. Write an expression for 'x minus 3.' ___ **x - 3**

3. Write an expression for 'twice a number n plus 5.' ___ **2n + 5**

4. Write an expression for 'half of t.' ___ **t/2**

Word problems

5. A taxi charges $3 plus $2 per mile. Write an expression for the cost of m miles.
Answer: 3 + 2m

6. A bag holds b apples. Write an expression for 5 bags plus 4 loose apples.
Answer: 5b + 4

Grade 6 Practice

Standard 6.EE.7

Computation

1. Solve: x + 5 = 12. x = ___ **7**

2. Solve: 3y = 18. y = ___ **6**

3. Solve: n - 4 = 10. n = ___ **14**

4. Solve: k/2 = 7. k = ___ **14**

Word problems

5. Sam has x dollars. He earns 9 more and now has 20 dollars. Find x.
Answer: 11 dollars

6. Each box holds b cards. 4 boxes hold 28 cards. Find b.
Answer: 7 cards

Grade 6 Practice

Standard 6.EE.8

Computation

1. Is 6 a solution to x > 5? Write yes or no. **yes**

2. Is 3 a solution to x < 3? Write yes or no. **no**

3. Which number satisfies x ≥ 4: 2, 3, or 4? Write the number. **4**

4. Write an inequality for 'y is less than 10.' ___ **y < 10**

Word problems

5. Max needs at least 25 points. Write an inequality for the points p he needs.
Answer: p ≥ 25

6. Jo has at most $15 to spend. Write an inequality for the amount s she can spend.
Answer: s ≤ 15

Grade 6 Practice

Standard 6.EE.9

Computation

1. In d = 5t, which variable is independent? Write d or t. **t**

2. In C = 3n, which variable is dependent? Write C or n. **C**

3. In the cost c = 4p, p is the ___ variable. **independent**

4. In y = x + 2, which variable is dependent? Write y or x. **y**

Word problems

5. A plant grows 2 cm per day. Its height h after d days is h = 2d. Which variable is independent?
Answer: d

6. Pay P depends on hours h worked: P = 12h. Which variable is dependent?
Answer: P

Grade 6 Practice

Standard 6.G.1

Computation

1. A triangle has base 6 and height 4. What is its area? ___ **12**

2. A parallelogram has base 5 and height 3. What is its area? ___ **15**

3. A triangle has base 10 and height 3. What is its area? ___ **15**

4. A trapezoid has bases 4 and 6 and height 3. What is its area? ___ **15**

Word problems

5. A triangular garden has a base of 8 m and a height of 5 m. What is its area?
Answer: 20 square meters

6. A parallelogram has base 7 and height 2. What is its area?
Answer: 14

Grade 6 Practice

Standard 6.G.2

Computation

1. A rectangular prism is 4 by 3 by 2. What is its volume? ___ **24**

2. A prism has a base area of 10 and a height of 5. What is its volume? ___ **50**

3. A cube has side length 3. What is its volume? ___ **27**

4. A prism has a base area of 6 and a height of 4. What is its volume? ___ **24**

Word problems

5. A box is 5 by 2 by 3 units. What is its volume?
Answer: 30 cubic units

6. A fish tank has a base of 4 by 6 feet and a height of 2 feet. What is its volume?
Answer: 48 cubic feet

Grade 6 Practice

Standard 6.G.3

Computation

1. A triangle has vertices (0, 0), (4, 0), and (4, 3). What is its area? ___ **6**

2. A rectangle has corners (1, 1), (5, 1), (5, 4), and (1, 4). What is its area? ___ **12**

3. What is the length of the segment from (2, 1) to (2, 7)? ___ **6**

4. What is the length of the segment from (-1, 3) to (4, 3)? ___ **5**

Word problems

5. A rectangle has vertices (0, 0), (6, 0), (6, 2), and (0, 2). What is its area?
Answer: 12 square units

6. Find the distance between the points (1, 2) and (1, 9).
Answer: 7 units

Grade 6 Practice

Standard 6.G.4

Computation

1. How many faces does a cube have? ___ **6**

2. How many edges does a rectangular prism have? ___ **12**

3. How many faces does a triangular prism have? ___ **5**

4. How many triangular faces does a square pyramid have? ___ **4**

Word problems

5. How many squares are in a net of a cube?
Answer: 6 squares

6. A net has 2 triangles and 3 rectangles. Which solid does it make?
Answer: triangular prism

Grade 6 Practice

Standard 6.SP.1

Computation

1. Which is a statistical question? Write A or B. A) How tall are students in our class? B) W

2. Is 'What is your favorite color?' a statistical question? Write yes or no. **yes**

3. Is 'How many pets do students in our class have?' a statistical question? Write yes or

4. Is 'What is 5 × 5?' a statistical question? Write yes or no. **no**

Word problems

5. Is 'Who is the tallest person in the room?' a statistical question? Write yes or no.
Answer: no

6. Is 'How long is a typical bus ride for our grade?' a statistical question? Write yes or no.
Answer: yes

Grade 6 Practice

Standard 6.SP.2

Computation

1. Data: 2, 4, 6, 8, 10. What is the median? ___ **6**

2. Data: 3, 3, 5, 9. What is the mode? ___ **3**

3. Data: 4, 9, 12, 20. What is the range? ___ **16**

4. Data: 1, 2, 3, 4. What is the median? ___ **2.5**

Word problems

5. Test scores are 70, 80, and 90. What is the mean?
Answer: 80

6. Heights are 50, 52, 54, and 56 inches. What is the range?
Answer: 6 inches

Grade 6 Practice

Standard 6.SP.3

Computation

1. Find the mean of 2, 4, and 6. ___ **4**

2. Find the median of 1, 3, 5, and 7. ___ **4**

3. Find the mode of 2, 2, 5, and 6. ___ **2**

4. Find the mean of 10, 20, 30, and 40. ___ **25**

Word problems

5. Sam's steps for 5 days were 3, 5, 7, 5, and 5. What is the mode?
Answer: 5 steps

6. Sam's test scores are 80, 90, and 100. What is the mean?
Answer: 90

Grade 6 Practice

Standard 6.SP.4

Computation

1. A dot plot shows 3 dots above the number 2. How many data points are there? ___

2. A histogram has 5 data points in the 0–4 bar. How many data points are there? ___

3. A dot plot has 2 dots at 1 and 3 dots at 2. How many data points in all? ___ **5**

4. A histogram has 6 data points in 10–20 and 4 in 20–30. How many data points in all?

Word problems

5. A dot plot shows 4 students with 2 pets. How many students are in the plot?
Answer: 4 students

6. A histogram has 8 data points in the 0–5 group and 2 in the 5–10 group. How many data points in all?
Answer: 10

Grade 6 Practice

Standard 6.SP.5

Computation

1. A data set has 8 values. How many values does it have? ___ **8**

2. Find the mean of 5, 7, and 9. ___ **7**

3. Find the range of 3, 8, and 10. ___ **7**

4. The measure of spread found by subtracting the minimum from the maximum is ca

Word problems

5. A data set has 6 values. The maximum is 20 and the minimum is 4. What is the range?
Answer: 16

6. Four values have a mean of 12. What is the sum of the four values?
Answer: 48